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Inventor(s)

Yeo; Terence et al.

MULTIFUNCTIONAL LIGHTING ASSEMBLY WITH FLEXIBLE PCB STRIP FOR A T-BAR GRID ARRANGEMENT

Abstract

A multifunctional lighting assembly is configured with an elongate body that serves as a cross tee within a T-bar grid arrangement and has at least one connecting member with a flexible PCB strip that provides bridging of a T-bar to make electrical connections between multifunctional support assemblies without need of cutting or significant displacement of ceiling tiles of the T-bar grid arrangement. Further embodiments provide for the support of ceiling panels within a suspended ceiling system, connection to a structural ceiling, and inclusion of further components such as light sources, optical components, sensors, electronic controllers, and wireless modules.

Inventors: Yeo; Terence (Concord, MA), Kelly; Timothy (Brookline, MA)

Applicant: Fusion Optix, Inc. (Woburn, MA)

Family ID: 83449927

Assignee: Fusion Optix, Inc. (Woburn, MA)

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to a support assembly for use in a suspended ceiling system. More particularly, the present disclosure relates to a multifunctional ceiling support assembly for use in a suspended ceiling system which typically supports ceiling tiles and lighting apparatuses.

BACKGROUND

[0002] Contemporary buildings, for example, houses or offices often employ a structural ceiling from which is supported a suspended ceiling arrangement. Typically, the suspended ceiling arrangement includes a plurality of ceiling tiles or panels hanging at a distance below the structural ceiling that may vary between only a few inches or several feet depending upon one or more constraints that are typically encountered in the space below the structural ceiling such as presence of HVAC ducting, electrical conduits, or a desired amount of height that should be made available for use in a room below the suspended ceiling arrangement.

[0003] In traditional configurations, the suspended ceiling arrangement further includes a plurality of T-bars that are configured to support the plurality of ceiling tiles or panels in position; the plurality of T-bars is suspended from the structural ceiling, for example via an arrangement of wires or rods. Specifically, such an arrangement of the plurality of T-bars provides cells to accommodate the plurality of ceiling tiles or panels therein. Additionally, a flush-finish of lower surfaces of the plurality of T-bars, and the plurality of ceiling tiles or panels are such that they appear as a continuous mono-planar lower ceiling surface. Typical common ceiling tiles sizes arranged in grid arrays include 12"×12", 12"×24", 24"×48", 24"×60", 24"×72", 30"×30", 20"×60", 30"×60", 60"×60", and 48"×48". Ceiling tiles for a "Lay-In" configuration typically have squared 90 degree edges while ceiling tiles for "Tegular" configurations are typically kerfed on the edges to allow the tiles to protrude slightly below the T-bars or a ceiling plane. Common standard widths of T-bar horizontal portions are 9/16", 15/16" and 1.5". These widths correspond with the apparent spacing between ceiling tiles as seen within a room.

[0004] Lighting devices or apparatuses are utilized in many diverse applications, such as in office workspaces, in hospitals, in warehouses, in educational institutions, in cleanrooms, in data centers, in research laboratories, in indoor and outdoor living spaces, in industrial areas, in vehicles and so forth to provide illumination for humans performing visual tasks. Contemporarily, lighting devices are also employed for aesthetic purposes to provide a visually comforting environment to a given person. In traditional configurations of installing these lighting devices, these lighting devices are known to have been affixed to ceilings, walls and other building elements to illuminate their environs.

[0005] In contrast to these highly traditional methods of installing the aforementioned lighting devices, some other traditionally configured suspended ceiling arrangements are also known to

have been additionally provided with lighting fixtures that are designed to replace 2×2 or 2×4 ceiling tiles in an arrangement that allows for operatively illuminating the surroundings, for example, a cubical space in an office, a corridor of a house, and the like. These light fixtures may also be designed to include as two or three fluorescent tubes as panels with light emitting diodes, for example, in the form of troffers that may require support on all four sides for accomplishing an installation of such light fixtures. Moreover, these lighting fixtures are arranged within the suspended ceiling arrangements to be supported at specific points, or in specific areas, of the suspended ceiling with an intention to achieve an aesthetically pleasant look. However, despite such intentions, the traditional configurations of such suspended ceiling arrangements are incapable of satisfying such desirable luminaires to meet aforementioned expectations.

[0006] Major issues that are encountered with the implementation of the traditionally configured suspended ceiling arrangements are that the traditionally configured suspended ceiling arrangements, during installation, are complex to retrofit in addition to being costly to replace during the retro-fitment process. Moreover, in use as a consequence to the implementation of these traditionally configured suspended ceiling arrangements itself, these traditionally configured suspended ceiling arrangements continue to offer a monotonous look and therefore, render the overall suspended ceiling with substantially unappealing aesthetics. On many occasions, an environment or workspace may also be additionally provided with multiple smaller lighting devices such as downlights that help provide an increased, focused, or in other words, specifically-directed light for allowing persons in the room to perform minutely detailed tasks. In many cases, these small lighting devices may include multiple light sources, for example, bulbs and the like. However, such use of multiple small light sources consequentially results in an increase in installation and maintenance costs besides leading to inefficient energy usage, wastage of resources and environmental pollution.

[0007] A further issue that is encountered with use of conventionally or traditionally designed suspended ceiling arrangements is that replacing the conventionally designed suspended ceiling arrangements, for example when refurbishing a given building, in which another conventionally or traditionally suspended ceiling arrangement previously exists, generates a lot of waste material that is potentially not straightforward to recycle or reuse. Moreover, such generation of waste material can be environmentally detrimental to achieving an energy efficient planet preferably with a minimal carbon-footprint.

[0008] Even further, a number of support elements, for example, the T-bars used together with the ceiling panels for forming the conventional suspended ceiling arrangements lack adequate provisions for enhancing the functionalities besides merely supporting the ceiling panels.

[0009] Therefore, taking aforementioned problems into consideration, there exists a need to overcome the aforementioned drawbacks associated with the existing T-bars and lighting devices used in conjunction with a suspended ceiling grid arrangement.

SUMMARY

[0010] The present disclosure seeks to provide a multifunctional ceiling support assembly for use in forming a suspended ceiling grid arrangement. The multifunctional ceiling support assembly is configured to be easier to manufacture, install and reconfigure after initial installation (for example to achieve one or more additional, or modified, functionalities). Further, the multifunctional ceiling support assembly is inexpensive to manufacture, i.e. owing to the simplified manufacturing and design process, easier to recycle or reuse when a building incorporating the suspended ceiling system is being dismantled or refurbished. Further, the present disclosure seeks to provide the multifunctional ceiling support assembly that is versatile in its use for providing an improved ease of control in the installation and use of light fixtures while also accommodating one or more concomitant power cable connections for rendering power supply to these light fixtures.

Furthermore, the present disclosure seeks to provide the multifunctional ceiling support assembly for use as a modular functional fixture that is capable of accommodating various other types of

user-operable objects such as optical elements, power modules, speakers, and the like.

Furthermore, the present disclosure seeks to provide a robust and flexible multifunctional ceiling support assembly that can be configured easily and quickly by a user during an installation process whilst suiting various requirements that may be encountered during the installation process.

Principally, the multifunctional ceiling support assembly of the present disclosure can function as a new T-bar (i.e. can replace conventional T-bar), can be used in conjunction with existing T-bar of the suspended ceiling grid arrangement, act as lighting device or apparatus, and enables in supporting ceiling tiles thereon. Also, the multifunctional ceiling support assembly provides versatility in terms of overall construction of the suspended ceiling grid arrangement, essentially in terms of lighting arrangement and supporting ceiling tiles.

[0011] In one aspect, the present disclosure provides a multifunctional ceiling support assembly for use in a suspended ceiling system having a T-bar grid arrangement supporting ceiling tiles. The multifunctional ceiling support assembly comprises an elongate body comprising an elongate planar base portion having first and second lateral ends, first and second longitudinal ends, and a top plane; first and second connecting members each positioned at the first and second longitudinal ends of the elongate body and configured with support portions capable of attachment to the T-bar grid arrangement; and a third connecting member mounted on the top plane of the elongate body, the third connecting extends up from the elongate body and functions to separate and position ceiling tiles within the T-bar grid arrangement.

[0012] Optionally, each of the first and second connecting members comprises a first support portion, a second support portion integral with the first support portion, and a third support portion integral with the second support portion. The first support portions of the first and second connecting members are adapted to be coupled to the top plane of the base portion at the first and second longitudinal ends, respectively, using fasteners. The second support portions of the first and second connecting members are configured to rest on horizontal legs of two laterally spaced apart T-bars. The third support portions are configured to be coupled with vertical legs of the two laterally spaced apart T-bars.

[0013] More optionally, each of the first and second connecting members further comprises a projecting tab integral with the third support portion, and wherein the projecting tab of each of the first and second connecting members is configured to be received by one vertical slot of a vertical leg of the two laterally spaced apart T-bars.

[0014] Optionally, the third connecting member comprises a first support portion coupled to the top plane of the base portion, and a second support portion extending vertically from the first support portion, the second support portion comprising a cut-out for receiving an anchor of a T-bar arranged between the two laterally spaced apart T-bars.

[0015] Optionally, the elongate body further comprises a first clip and a second clip extending generally upwardly from the top plane at the first and second lateral ends, respectively, of the base portion, wherein the first clip, the second clip and the top plane of the base portion define a first channel therebetween, and a first protrusion and a second protrusion extending generally downwardly from the bottom plane at the first and second lateral ends, respectively, of the base portion, wherein the first protrusion, the second protrusion and the bottom plane of the elongate planar base portion define a second channel therebetween.

[0016] Optionally, each of the first, second and third connecting members further comprises an anchoring hole that serves as an attachment point for suspension from a structural ceiling.

[0017] Optionally, the multifunctional support assembly is operable to connect two T-bars within the suspended ceiling system.

[0018] Optionally, the elongate body houses a light source.

[0019] More optionally, the light source comprises a light emitting diode.

[0020] Optionally, a width of the elongate body is less than or equal to the width a T-bar within the T-bar grid arrangement.

[0021] More optionally, the elongate body is at least partially recessed within tegular edges of two abutting ceiling tiles.

[0022] More optionally, a light source is recessed within the tegular edges of two abutting ceiling tiles.

[0023] Optionally, a ceiling tile rest upon the top plane of the elongate body.

[0024] Optionally, a width of the elongate body is greater than the width of a T-bar within the T-bar grid arrangement.

[0025] Optionally, the elongate body is oriented at an angle not parallel or perpendicular to the T-bar grid arrangement.

[0026] Optionally, the elongate body houses a reflector.

[0027] Optionally, the multifunctional support assembly further comprises a light guide having an input edge within the elongate body.

[0028] Optionally, the multifunctional ceiling support assembly further comprises a light source and an upward projecting lens suspended below the elongate body.

[0029] Optionally, the multifunctional ceiling support assembly further comprises a component selected from a group consisting of an alarm, sensors, ventilation fan, heater, humidifier, electronic controller, power supply, battery, wireless communication module, light guide, reflector.

[0030] Optionally, the multifunctional support assembly of claim **1**, further comprising mechanical features, electrical connectors, etc.

[0031] Additional aspects, advantages, features and objects of the present disclosure would be made apparent from the drawings and the detailed description of the illustrative embodiments construed in conjunction with the appended claims that follow.

[0032] It will be appreciated that features of the present disclosure are susceptible to being combined in various combinations without departing from the scope of the present disclosure as defined by the appended claims.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0033] The summary above, as well as the following detailed description of illustrative embodiments, is better understood when read in conjunction with the appended drawings. For the purpose of illustrating the present disclosure, exemplary constructions of the disclosure are shown in the drawings. However, the present disclosure is not limited to specific methods and instrumentalities disclosed herein. Moreover, those in the art will understand that the drawings are not to scale. Wherever possible, like elements have been indicated by identical numbers.

[0034] Embodiments of the present disclosure will now be described, by way of example only, with reference to the following diagrams wherein:

[0035] FIG. **1A** (Prior Art) is a perspective top view of a T-bar of conventional design, in accordance with an embodiment of the present disclosure;

[0036] FIG. **1B** (Prior Art) is a perspective bottom view of a conventional T-bar ceiling grid system, in accordance with an embodiment of the present disclosure;

[0037] FIG. **2** is a perspective top view of a multifunctional ceiling support assembly, in accordance with an embodiment of the present disclosure;

[0038] FIG. **3** is a transverse cross-section view of the multifunctional ceiling support assembly of FIG. **2** with ceiling tiles of tegular design being supported by the multifunctional ceiling support assembly, in accordance with an embodiment of the present disclosure;

[0039] FIG. **4** is a perspective top view of a multifunctional ceiling support assembly, in accordance with another embodiment of the present disclosure;

[0040] FIG. **5** is a perspective top view of a multifunctional ceiling support assembly, in

accordance with yet another embodiment of the present disclosure;

[0041] FIG. **6** is a perspective top view of the multifunctional ceiling support assembly of FIG. **2** coupled to a pair of T-bars offset along the longitudinal axis of the multifunctional ceiling support assembly, in accordance with yet another embodiment of the present disclosure;

[0042] FIG. **7A** is a top perspective view of a multifunctional ceiling support assembly having an edge lit light guide, in accordance with yet another embodiment of the present disclosure;

[0043] FIG. **7B** is a transverse cross-section view of the multifunctional ceiling support the assembly of FIG. **7A** with ceiling tiles resting thereon, in accordance with an embodiment of the present disclosure;

[0044] FIG. **8** is longitudinal cross-section view of the multifunctional ceiling support assembly of FIG. **6** with lay-in type ceiling tiles supported thereon, in accordance with an embodiment of the present disclosure;

[0045] FIG. **9** is a longitudinal cross-section view of the multifunctional ceiling support assembly of FIG. **4**, in accordance with an embodiment of the present disclosure;

[0046] FIG. **10A** is a bottom perspective view of a multifunctional ceiling support assembly wherein an elongate body houses an optical arrangement having collimating reflector lenses, in accordance with an embodiment of the present disclosure;

[0047] FIGS. **10B-10D** are perspective bottom views of a multifunctional ceiling support assembly with differing optical arrangements housed within the elongate body, in accordance with various embodiments of the present disclosure;

[0048] FIG. **11** is a top perspective view of a multifunctional ceiling support assembly having a downward projecting light guide that is mounted parallel with a longitudinal axis of the multifunctional ceiling support assembly, in accordance with an embodiment of the present disclosure;

[0049] FIG. **12** is a top perspective transverse cross-section view of a multifunctional ceiling support assembly having a light guide within the elongate body that is parallel with the general ceiling plane and extending along the longitudinal axis of the multifunctional ceiling support assembly, in accordance with an embodiment of the present disclosure;

[0050] FIG. **13** is a top perspective view of a multifunctional ceiling support assembly having an upward projecting optical arrangement that is positioned below the elongate body and mounted parallel with the longitudinal axis of the multifunctional ceiling support assembly, in accordance with an embodiment of the present disclosure;

[0051] FIGS. **14A-14F** show in polar plot form differing light distribution patterns that can be achieved with various embodiments of particular optical arrangement, in accordance with various embodiments of the present disclosure;

[0052] FIGS. **15A-15B** show multifunctional ceiling support assembly embodiments with alternative arrangements of sensors and electronics housed within elongate bodies, in accordance with various embodiments of the present disclosure;

[0053] FIG. **16** is a perspective top view of a multifunctional ceiling support assembly showing an exemplary configuration of at least one of end caps for facilitating a wiring connection therethrough, in accordance with an embodiment of the present disclosure;

[0054] FIGS. **17A-17G** are schematic top view of multifunctional ceiling support assembly showing other exemplary configurations of end caps for facilitating the wiring connection therethrough, in accordance with various embodiments of the present disclosure;

[0055] FIG. **18** is a perspective top view of the multifunctional ceiling support assembly from FIG. **16** used with a flexible PCB, in accordance with an embodiment of the present disclosure; and

[0056] FIG. **19** is a bottom view of a suspended ceiling grid system layout wherein multifunctional ceiling support assemblies are mounted non-parallel to the ceiling grid T-bars, in accordance with an embodiment of the present disclosure.

[0057] In the accompanying drawings, an underlined number is employed to represent an item over

which the underlined number is positioned or an item to which the underlined number is adjacent. A non-underlined number relates to an item identified by a line linking the non-underlined number to the item. When a number is non-underlined and accompanied by an associated arrow, the non-underlined number is used to identify a general item at which the arrow is pointing.

DETAILED DESCRIPTION

[0058] The following detailed description illustrates embodiments of the present disclosure and ways in which they can be implemented. Although some modes of carrying out the present disclosure have been disclosed, those skilled in the art would recognize that other embodiments for carrying out or practicing the present disclosure are also possible.

[0059] In overview, embodiments of the present disclosure are concerned with a multifunctional ceiling support assembly that functions within a suspended ceiling system to support and align one or more ceiling tiles, connect one or more T-bars or other structural grid elements, and provide a means of connecting suspension cables or other suspension hardware.

[0060] Modular ceiling systems are employable to implement suspended ceilings, also referred to herein as “suspended ceiling grid arrangements”. Herein, the term “suspended ceiling grid arrangements” refers to a ceiling including a ceiling grid suspended or hung at a height below a structural ceiling of a given architecture, such as a room of a house, or a building. It will be appreciated that the structural ceiling is an overhead interior surface that covers, and defines, a maximum upper spatial limit of a room. In an example, the structural ceiling may be at a height of 2.5 meters from a floor (not shown) of the room. In such an example, the height below the structural ceiling for holding the suspended ceiling grid arrangement is 0.25 (approximately) metres (meters) from the height of the structural ceiling, i.e. 2.25 (approximately) metres (meters) from a floor of the room. Furthermore, the suspended ceiling grid arrangement is suspended or hung at the height using wires that hang from, and are securely fixed to, the structural ceiling. Optionally, the hanging wires can be hinged, hooked, tied, coupled, plastered securely or affixed to the structural ceiling. In an instance, during installation, the hanging wires are coupled to the structural ceiling to support the suspended ceiling grid arrangement to be hung at the desired height therefrom. Furthermore, the suspended ceiling grid arrangement is supported by the hanging wires at the desired height to provide a gap between the structural ceiling and the suspended ceiling grid arrangement. Beneficially, the gap provides a space for other electrically and/or electronically operated devices to be arranged therebetween.

[0061] Typically, the suspended ceiling grid arrangement includes a grid formation constructed using metallic bars (T-bars). The term “T-bars” used herein relates to hardware components such as an elongate rigid spine extending between mutually opposing walls that form terminal ends of the ceiling. Additionally, the T-bars include an inverted T-shaped structure comprising a flat vertical leg (supporting portion) integral to a flat horizontal leg (base portion). Furthermore, the T-bars include either a fixed anchor or an adjustable anchor for attachment to an adjacent member, such as another T-bar(s) or other holding arrangement(s) for securely holding or suspending the T-bar. The T-bars are of various styles, types and sizes and are generally characterized by the grid type. The term “grid type” refers to a property of the T-bar indicating the type, style or size of the T-bar being implemented. Generally, the grid type of the T-bar is a 15/16” flat style. However, other different grid types or other styles may also be implemented without limiting the scope of the disclosure. For instance, other grid type includes, but is not limited to, a slimline grid type or 9/16” flats including a 9/16” slot, a concealed grid type or a semi-concealed grid type. Moreover, larger T-Bars including 1.5” and 2” grid type are employed in industrial applications such as clean rooms, data centers, food processing plants and so forth. Optionally, the T-bars are conjoined to the hanging wires, either by hooking, welding, gluing, and so forth. Moreover, the T-bars include tracks or holes to which the hanging wires can be coupled to and/or can be latched onto for supporting (i.e. holding or suspending) the suspended ceiling grid arrangement from the structural ceiling. Furthermore, the T-bars of the suspended ceiling grid arrangement form an array of cells into which ceiling panels

can be arranged. Moreover, the array of cells is formed by the grid of horizontal legs of the T-bars. Optionally, the T-bars also include axes (imaginary line) passing through joining of the vertical legs and the horizontal legs. Furthermore, the axes cross each other when two T-bars overlap or cross each other either perpendicularly or at an angle.

[0062] Furthermore, the term “ceiling panels” as used herein relates to a lightweight structure, usually a shallow “cuboidal” structure, having a length, a breadth, and a height which are placed within an opening formed by the T-bars for providing a planar lower surface of the suspended ceiling. Optionally, the ceiling panels are fabricated from a porous cellular structure, having gaseous voids therein. Optionally, the ceiling panels are implemented as a plurality of substantially mutually identical panels, wherein each panel has major exterior surfaces that are substantially rectangular in form, for example square in form, when viewed from the given room. In some cases, the ceiling panels may have edges that are adapted to include one or more edgewise protruding lips and/or define one or more edgewise grooves i.e., along a length of the edge for enabling the ceiling panels to be securely held, or supported, during installation onto the grid.

[0063] Furthermore, the grid formation is configured to accommodate various electronic and/or electrical devices for providing a plurality of services in the room. Examples of various electronic and electrical devices may include at least one of: lights, alarms, sensors, ventilation fans, heaters, humidifiers, electronic controller, power supply, battery, wireless communication module, light guide, reflector and the like. For example, the suspended ceiling grid arrangement can include a power system for supplying electric power to the various electrically and/or electronically operated devices.

[0064] FIG. 1A is a perspective top view of a typical T-bar **10**, of conventional design and FIG. 1B is a bottom view of the same type of typical T-bar shown in a suspended ceiling grid arrangement. The T-bar **10** includes a vertical leg **12** and a horizontal leg **14**. In a suspended ceiling grid arrangement, the vertical leg **12** of each T-bar **10** has one or more holes **16** periodically passing therethrough through which a wire or any other suspension element may connect with and support the suspended ceiling array of T-bars at the desired elevation above the floor. Further, the T-bar **10** includes one or more slots **18** configured to accommodate an anchor **22** of the T-bar **10**. Furthermore, the horizontal legs **14** of the T-bars **10** define the general ceiling plane for supporting the plurality of ceiling panels. The general ceiling plane is formed by the horizontal legs **14** of the T-bar **10**. Herein, the axes of the horizontal legs **14** of the T-bars **10** are coplanar to the general ceiling plane. Additionally, the general ceiling plane is parallel to the floor of, for example, a room of a house, office, hallway, clean room etc. Optionally, the general ceiling plane is at a height 2.5 to 3 metres above the floor of the house. Typically, the vertical portion **12** of the T-bar **10** comprises the anchor **22** configured to be mechanically coupled with other mechanical components, the wall or ceiling to support a multifunctional ceiling support assembly. FIG. 1B shows a ceiling tile **20** installed in the suspended ceiling grid arrangement whereby the edges are resting upon horizontal legs **14** of T-bars.

[0065] FIG. 2 is a perspective top view of a multifunctional ceiling support assembly **100** that can be replaced by or used in conjunction with the T-bar **10** explained in conjunction with the FIGS. 1A-B. Further, FIG. 3 illustrates is a transverse cross-section view of the multifunctional ceiling support assembly of FIG. 2. The multifunctional ceiling support assembly **100** includes the elongate body **102**. Typically, the elongate body **102** is oriented parallel to the T-bar grid arrangement, i.e. oriented parallel to the horizontal legs **14** of T-bars **10**. In an embodiment, the the elongate body **102** may be oriented at an angle not parallel or perpendicular to the T-bar grid arrangement. In an example, longitudinal ends of the elongate body **102** may be distant from the T-bar grid arrangement in a non-uniform manner. Alternatively, the elongate body **102** may be configured to have a non-uniform height at the longitudinal ends thereof. The elongate body **102** is in this embodiment an extruded profile shape, typically comprised of metal, polymer, or composite material. Further, the elongate body **102** includes an elongate planar base portion **108** (hereinafter

referred to as ‘the base portion’). The base portion **108** is fabricated in a shape of a flat shaped member having a top plane **110** and a bottom plane **112**. It will be seen that the top plane **110** and the bottom plane **112** form two opposing sides of the base portion **108**. The base portion **108** of the elongate body **102** has a first lateral end **114**, a second lateral end **116**, a first longitudinal end **117**, and a second longitudinal end **119**.

[0066] Furthermore, as illustrated in FIG. 3, the elongate body **102** includes a first clip **118** extending generally upwardly from the top plane **110** of the base portion **108**. The first clip **118** is generally extending from the first lateral end **114** of the base portion **108**. Moreover, the elongate body **102** includes a second clip **120** extending generally upwardly from the top plane **110** of the base portion **108**. The second clip **120** is generally extending from the second lateral end **116** of the base portion **108**. As shown, the first clip **118** and the second clip **120**, along with the base portion **108**, define a first channel **122** therebetween. The first channel **122** is in a form of a hollow space, above the base portion **108**, adapted to receive, for instance, a first support portion **162** of a third connecting member **160**. The first and second clips **118**, **120** of the elongate body **102** are adapted to co-operatively secure the third connecting member **160** within the first channel **122**. It will be appreciated that a width of the base portion **108** is beneficially defined as such to match the horizontal width of the T-bars within the suspended ceiling grid system. For example, in an embodiment, the width of the elongate body **102**, particularly, the base portion **108**, is less than or equal to the width a T-bar within the T-bar grid arrangement. In another embodiment, the width of the elongate body **102** is greater than the width of a T-bar within the T-bar grid arrangement.

[0067] Furthermore, as illustrated in FIG. 3, the elongate body **102** includes a first protrusion **124** extending generally downwardly from the bottom plane **112** of the base portion **108**. The first protrusion **124** is generally extending from the first lateral end **114** of the base portion **108**. Moreover, the elongate body **102** includes a second protrusion **126** extending generally downwardly from the bottom plane **112** of the elongate body **102**. The second protrusion **126** is generally extending from the second lateral end **116** of the base portion **108**. As will be seen, the first and second protrusions **124**, **126** together with the base portion **108** define a second channel **128** (interchangeably referred to as void space) therebetween. The second channel **128** is in a form of a hollow space, below the base portion **108**, adapted to receive the at least one elongate light assembly **104** in the multifunctional ceiling support assembly **100**. As shown in FIG. 3, the elongate body **102** houses at least one elongate light assembly **104**. Herein, the elongate light assembly **104** includes a printed circuit board (PCB) **106** and one or more light emitting diodes (LEDs) **107** (only one being shown in the illustration) arranged on the PCB **106**. Typically, the LEDs **107** are arranged on the PCB **106** in a predefined pattern to provide a desired pattern of illumination. The elongate light assembly **104** is arranged and supported in the elongate body **102**, in the ceiling support assembly **100**. It will also be appreciated that although the elongate light assembly **104** has been shown and described to include the PCB **106** with the plurality of light emitting diodes **107** arranged thereon, the elongate light assembly **104** optionally has any other suitable configuration without departing from the scope and the spirit of the present disclosure. Further, the second is beneficially defined to allow for accommodation of the PCB **106** of the elongate light assembly **104** in the multifunctional ceiling support assembly **100**. In this manner, the elongate light assembly **104** is arranged in the second channel **128** of the elongate body **102**. It will be appreciated from the illustration of FIG. 3 that the second channel **128** in the elongate body **102** provides a space to accommodate the elongate light assembly **104** assembly of the multifunctional ceiling support assembly **100**.

[0068] As illustrated in FIG. 3, the elongate light assembly **104** includes at least one optical element **136** supported in the second channel **128**. In the present examples, the optical element **136** is beneficially a lens that, for example, may rely on the principle of total internal reflection or include a rotationally symmetrical design for providing a uniform light distribution. Further, as illustrated in FIG. 3, each of the first and second protrusions **124**, **126** comprise inwardly projecting

lips **132**

[0069] for slidably supporting the optical element **136** within the second channel **128**. Moreover, as illustrated in FIG. 3, the elongate light assembly **104** includes at least one reflector **138** arranged in the second channel **128**. The at least one reflector **138** is positioned around the one or more LEDs **107** (light emitting diodes) and located above the optical element **136**.

[0070] By accommodating the elongated light assembly **104**, comprising the PCB **106**, the LEDs **107** and the optical element **136** in a compact manner between the first and second protrusions **124**, **126**, i.e., within the second channel **128** of the elongate body **102**, an amount of complexity in the design of the multifunctional ceiling support assembly **100** is simplified to a minimum while an amount of time and effort that is incurred during installation of the ceiling support assembly **100** is also reduced. With continued reference to, and as shown in the view of, FIG. 3, the multifunctional ceiling support assembly **100** may further comprise a sensor arrangement to provide input for controlling operation of the elongate light assembly **104**. The sensor arrangement may be configured to switch the elongate light assembly **104** based on at least a user requirement or a user activity. The term “user activity” refers to a movement or action detected by at least one of the sensors in the sensor arrangement, based on which the elongate light assembly **104** is configured to behave, or operate, corresponding to the user activity. In an implementation scenario, the sensor arrangement may optionally include an Internet-of-Things (IoT) device that is configured for switching the elongate light assembly **104** “ON” or “OFF”. Moreover, the elongate light assembly **104** may be configured to detect a presence or movement of a user for switching the lighting arrangement “ON” or “OFF”. In an example, user activity may include a hand clap, hand movement, finger pinch or flick and the like.

[0071] Further, any type of sensors may be implemented for use within the sensor arrangement. For example, the sensors forming part of the sensor arrangement may include one or more of an audio sensor, acoustic sensor, motion sensor, position sensor, proximity sensor, light sensor, ultrasonic sensor and accelerometer. In an example, the sensor arrangement may also be equipped with a sound recognition sensor (e.g. an ultrasound sensor) configured to control the elongate light assembly **104** via another device, such as an ultrasonic hand controller, a near-field wireless device or even ordinary acoustic signals such as human clapping of hands.

[0072] Referring now to FIGS. 2, 4-6 and 8, in embodiments herein, the multifunctional ceiling support assembly **100** also includes first and second end caps **130** that are arranged on respective first and second longitudinal ends **117**, **119** of the elongated body **102**.

[0073] As shown best in the view of FIG. 2, the ceiling support assembly comprises first and second connecting members **134a**, **134b**. Each of the first and second connecting members **134a**, **134b** includes a first support portion **140**, a second support portion **142**, and a third support portion **144**. The second support portion **142** is integral with the first support portion **140** and the third support portion **144** is integral with the second support portion **142**. As shown, the second support portion **142** extends linearly from the first support portion **140** while also being oriented orthogonally to the third support portion **144**.

[0074] The first support portions **140** of the first and second connecting members **134a**, **134b** are adapted to be coupled to the top plane **110** of the base portion **108** at the first and second longitudinal ends **117** and **119** respectively using fasteners **146**. Further, the second support portions **142** of the first and second connecting members **134a**, **134b** are configured to rest on horizontal legs **14** of two laterally spaced apart T-bars **10b**, as shown in the view of FIG. 6. The third support portions **144** enables in mounting the multifunctional ceiling support assembly **100** between two laterally spaced apart T-bars **10b**. Furthermore, the third support portions **144** are configured to be coupled with vertical legs **12** of the two laterally spaced apart T-bars **10b** by use of connecting anchors **22**, as shown in FIG. 4.

[0075] Moreover, as shown in the view of FIG. 2, each of the first and second connecting members **134a**, **134b** further comprises a projecting tab **148** integral with the third support portion **144**. The

projecting tab **148** of each of the first and second connecting members **134a**, **134b** is configured to be received by vertical slots **18** defined in the vertical legs **12** of the two laterally spaced apart T-bars **10b**, as shown in FIG. **6**.

[0076] Each of the first and second connecting members **134a**, **134b** also includes a support tab **150** integral with the third support portion **144**. As shown in the view of FIG. **2**, this support tab is configured to define a hole **152** for supporting a suspension cable therethrough. In other words, the first and second connecting members are configured with support portions, such as the projecting tab **148** and the support tab **150**, capable of attachment to the T-bar grid arrangement.

[0077] Further, as shown in FIGS. **4** and **5**, each of the third support portions **144** of the first and second connecting members **134a**, **134b** comprises a hook portion **154**. The hook portion **154** of each of the first and second connecting members **134a**, **134b** is configured to be mounted on one of the vertical legs **12** of the two laterally spaced apart T-bars **10b**.

[0078] As shown best from the views of FIGS. **3** and **8**, the first and second clips **118**, **120** on the base portion **108** and the second support portions **142** of the first and second connecting members **134a**, **134b** are configured to support the ceiling tiles **22** thereon.

[0079] Additionally, turning back to, and as shown best in, the view of FIG. **4**, each of the third support portions of the first and second connecting members **134a**, **134b** further comprises a cut-out **156**. The cut-outs **156** of the first and second connecting members **134a**, **134b** are configured to receive anchors **22** of the two T-bars **10a** arranged longitudinal to the multifunctional ceiling support assembly **100**, as also shown in the view of FIG. **9**. Alternatively, as shown in the view of FIG. **5**, the second support portion **142** may also be configured to extend orthogonally and centrally from both the first and third support portions **140**, **142**. In this alternative configuration, the first and third support portions **140**, **142** would also be spaced apart and oriented orthogonally to each other. Further, the second support portion **142** defines a hole **158** for supporting the same, or another, suspension cable therethrough.

[0080] Further, as shown in FIGS. **2**, **3** and **6**, the multifunctional ceiling support assembly **100** may additionally include a third connecting member **160** that can be used, depending on the specific application, for any combination of 1) connecting the multifunctional ceiling support assembly with a T-bar, 2) serving as a suspension attachment location, and 3) serving as a positioning bracket to hold a ceiling tile in place and prevent excessive lateral movement. The third connecting member **160** extends up from the elongate body **102** and functions to separate and position ceiling tiles **20** within the T-bar grid arrangement. The third connecting member **160** includes a first support portion **162** and a second support portion **164**.

[0081] As shown in the views of FIGS. **3** and **6**, the first support portion **162** is coupled to the top plane **110** of the base portion **108**. Further, the second support portion **164** extends vertically from the first support portion **162**. Moreover, the second support portion **164** defines a cut-out **166** for receiving the anchor **22** of a T-bar, for example, an intermediary T-bar (not shown) that may be arranged between the two laterally spaced apart T-bars **10b**. Furthermore, the second support portion also defines one or more holes **168** for supporting a suspension cable therethrough. Additionally, the third connecting member **160** serves as a positioning spacer to hold a ceiling tile in place and prevents excessive movement in a direction transverse to the multifunctional ceiling support assembly.

[0082] By providing one or more holes **168** in the second support portion **164** of the third connecting member **160**, the multifunctional ceiling support assembly **100** is rendered with improved versatility to accommodate for various spatial adjustments, for example, in cases where tolerances would have otherwise been needed to accommodate receipt of the suspension cable therethrough.

[0083] Accordingly, from the foregoing description, it is evident that the multifunctional ceiling support assembly **100** of the present disclosure serves to support and align one or more ceiling tiles **20** while connecting one or more T-bars in the grid. Typically, the ceiling tile **20** rest upon the top

plane **110** of the elongate body **102**. Further, the multifunctional ceiling support assembly **100** also provides a means of connecting the suspension cables or other similar, or dissimilar, suspension hardware thereto. In other words, each of the first, second and third connecting members **134a**, **134b**, **160** further comprises an anchoring hole, such as the holes **152**, **168** that serves as an attachment point for suspension from a structural ceiling.

[0084] Referring now to FIG. 7A (top perspective view) and FIG. 7B (transverse cross-section view) are illustrations of a multifunctional ceiling support assembly embodiment comprising a double edge lit light guide **137C**. The dotted line and “7B section” notation with arrow in FIG. 7A identify the cross-section orientation of the FIG. 7B view. The light guide **137C** is mounted inside an elongate body **702**. The elongate body **702** is wider than the T-bar horizontal leg **14** and mounted flush with the underside of the ceiling panel **20**. This embodiment provides a unique form factor with aesthetic and functional advantages. The increased width allows for a wider light guide and therefore increased light emitting surface. This can be useful in reducing peak brightness and glare. The shallow height of the elongate body **702** minimizes protrusion into the illuminated space below. FIG. 7B shows Lay-In type ceiling tiles **20** being positioned in place by the third connecting member **160** which also serves as an available attachment point for suspension connection to the structural ceiling. LEDs **107**, are typically arranged in linear arrays to input light into one or two edges of the light guide **137C**.

[0085] FIG. 8 is longitudinal cross-section view of the multifunctional ceiling support assembly **100** of FIGS. 2, 3 and 6 mounted in a suspended ceiling system. The ceiling tile **20** is positioned in a Lay-In type configuration with a 90 degrees squared edge that positions the ceiling tile **20** directly upon the top plane **110** of the base portion **108** as well as on the second supporting section **142** and the third connecting member **160**. The projecting tab **148** engages with the vertical portion **12** of the T-bar **10** to secure the position of the second supporting section **142** which rests upon the horizontal leg **14** of the T-bar **10**. The support tab **150** includes the hole **152** which serves an attachment point for means of suspension such as for example, a suspension cable.

[0086] FIG. 9 is a longitudinal cross-section view of the multifunctional ceiling support assembly **100** embodiment of FIG. 4 wherein the hook portion **154** of the connecting member latches over the vertical portion of the T-bar **10**. In other words, the multifunctional support assembly **100** is operable to connect two T-bars **10** within the suspended ceiling system.

[0087] FIG. 10A is a bottom perspective view of a multifunctional ceiling support assembly embodiment of FIG. 2 installed in a suspended ceiling grid system. The ceiling tiles **20** are of tegular type with partially exposed edges **21** which extend below the T-bar horizontal leg **14**. The elongate body **102** houses an optical arrangement comprising reflector lenses **111** which collimate light from LED **107** light sources. The end cap **130** serves as a first support section **140** of the connecting member which attaches to the T-bar grid.

[0088] FIGS. 10B-10D are perspective bottom views of a multifunctional ceiling support assembly embodiments with differing optical arrangements housed within the elongate body **102**. FIG. 10B is an embodiment wherein the optical element **136B** is a translucent lens that is mounted flush with a bottom plane of the elongate body **102**. FIG. 10C is an embodiment wherein the optical element **136C** is a translucent lens that is mounted recessed within the elongate body **102**. This can be advantageous in some applications where reducing glare is important. FIG. 10D is an embodiment which includes black baffling **113** around reflector optics as means for reducing glare.

[0089] FIG. 11 is a top perspective view of the multifunctional ceiling support assembly embodiment of FIG. 2 further comprising a light guide **137A** that is mounted parallel with the longitudinal axis of the multifunctional ceiling support assembly and projects in a downward direction. Light is input into an input edge of the light guide **137A** by LEDs within the light elongate body **102** and light is propagated downward by internal reflection. Means for extraction of light include surface features on the external surface of the light guide as well as light scattering materials within the volume of the light guide.

[0090] FIG. 12 is a top perspective transverse cross-section view of a multifunctional ceiling support assembly embodiment further comprising an edge lit light guide 137B within the elongate body that is parallel with the general ceiling plane and extending along the longitudinal axis of the multifunctional ceiling support assembly. The light guide 137A includes an input edge within the elongate body 102. The ceiling tiles 20 are of tegular type with kerfed edges that provide a cavity for the elongate body 102 to fit within and allow the option of mounting flush with the bottom of the ceiling tiles 20. The third connecting member 160 extends up between the ceiling panels to hold ceiling tiles in position and serve as an optional suspension anchor point. In this particular embodiment, LEDs 107 are mounted horizontally along a single edge which can be utilized to achieve asymmetric output. Also, as shown in FIG. 12, the elongate body 102 is at least partially recessed within tegular edges of two abutting ceiling tiles 20. Further, the light source (or edge lit light guide 137B) is also recessed within the tegular edges of two abutting ceiling tiles 20. As shown in FIG. 12, lower surfaces of the light guide 137A and the two abutting ceiling tiles 20 conform (i.e. on a same plane), however, in another embodiments the lower surface of the light guide 137A may rest inside a recess of the abutting ceiling tiles 20 or may project outside the recess of the abutting ceiling tiles 20. It will be evident that based on a height of the elongate body 102, the lower surface of the light guide 137A may rest inside or project outside the recess.

[0091] FIG. 13 is a top perspective view of a multifunctional ceiling support assembly embodiment further comprising an upward projecting optical arrangement that is positioned below the elongate body and mounted parallel with the longitudinal axis of the multifunctional ceiling support assembly. A lens 139 is positioned over a series of LEDs 107 and used to project light in an upward direction onto the bottom surface of ceiling tiles 20 from where it is subsequently reflected downward into an illuminated space below. In this indirect lighting configuration the lens 139 can be optimized to provide a uniform light pattern upon the bottom surface of the ceiling tiles 20 which are typically white and of high reflectance. A suspension cable 141, rod or bracket is used to position the lens 139 below the elongate body 102 and connects from the elongate body 102 to a lens connecting member 143 which supports the lens 139 and light source, LEDs 107. A third connecting member 160 extends up between ceiling panels to hold ceiling tiles 20 in position and serve as an optional suspension anchor point to the structural ceiling.

[0092] FIGS. 14A-14F show in polar plot form differing light distribution patterns that can be achieved with various embodiments of multifunctional ceiling support assembly embodiments having particular optical arrangements. All figures are polar plots of relative intensity vs. angle of light distribution. FIG. 14A shows a Lambertian distribution which can be achieved for example with a diffuse optical element implemented in the configuration of FIG. 3. FIG. 14B and FIG. 14C show medium and narrow collimated beams respectively. These can be achieved, for example, with optimization of reflectors shown in FIG. 3 and FIG. 10. FIG. 14D shows a narrow asymmetric light distribution which can be achieved, for example, by optimization of the reflector and optical element of FIG. 3. FIG. 14E shows a wall wash light distribution which can be achieved, for example, by optimization of the embodiment of FIG. 12. FIG. 14F shows a wide spread light distribution which can be achieved, for example, by optimization of the embodiment of FIG. 11. The specific light distributions and means for achieving them are representative but not comprehensive examples.

[0093] FIG. 15A is a top perspective view which shows a multifunctional ceiling support assembly embodiment with a sensor 151 housed within the void space 128 of the elongate body 102. A sensor head 153 extends below the elongate body 102 to receive stimulus from the environment, for example, ambient light level or motion detection. Additional electronics such as electronic controllers, wireless communication components, power supplies, and batteries may also be housed within the elongate body. Lay-In type ceiling tiles 20 are held in position by the third connecting member 160 which also serves as an optional attachment point for suspension from the structural ceiling.

[0094] FIG. **15B** is a top perspective view which shows a multifunctional ceiling support assembly embodiment which includes a power supply **143** and an edge lit light guide **137B** housed within the void space **128** of the elongate body **102**. Lay-In type ceiling tiles **20** are held in position by the third connecting member **160** which also serves as an optional attachment point for suspension from the structural ceiling.

[0095] FIG. **16** illustrates a perspective top view of the multifunctional ceiling support assembly **1000** showing an exemplary configuration of at least one of the end caps, for instance, the end cap **1030a** that facilitates a wiring connection **1072** therethrough. It is to be noted that the term ‘wires’ as disclosed herein is used to reference singular strands of wires, up to, multi-cored or multi-strand wires together with their respective insulation sheathings, or layers, as would be encountered in the case of flat profile cables.

[0096] Moreover, as shown in the view of FIG. **16**, at least on one of the first and second connecting members (for instance, the first connecting member **1034a** corresponding to the first longitudinal end **117**), the first end cap **1030a** may be configured to terminate partway, along a length of, the second support portion **1042** itself so as to form a ledge. In the view of FIG. **16**, it may be noted that the second support portion does not extend into the third support portion, for example, the third support portion **144** that is instead shown in the views of FIGS. **2**, **4-6** and **8-9** respectively.

[0097] FIGS. **17A-17G** are schematic top view of the multifunctional ceiling support assembly **1100**, **1102**, **1104**, **1106**, **1108**, **1110**, **1112** showing other exemplary configurations of end caps for facilitating the wires **1172** therethrough. Also, as shown by way of FIGS. **17A** through **17G**, the elongated body **1102** may have different configurations of the second support portions **1142a** through **1142g** that can be contemplated, in lieu of that disclosed in conjunction with FIG. **16**. Also, from the views of FIGS. **17A** to **17G**, it is also evident that a shape of the second support portion **1142** is changed to suit specific requirements dictated by a specific type of suspended ceiling grid application, for example, when the second support portions **1142** are adapted to meet with a specific size, shape, and/or configuration of the T-bars **10** that are being used in forming the suspended ceiling grid. In an example as shown in the view of FIG. **17A**, the second support portion **1142a** is of a quadrilateral configuration.

[0098] In another example as shown in the view of FIG. **17B**, the second support portion **1142b** is of a semi-hexagonal configuration. In yet another example as shown in the view of FIG. **17C**, the second support portion **1142c** is of a semi-circular configuration. In yet another example as shown in the view of FIG. **17D**, the second support portion **1142d** is of a broadened-butt configuration. In yet another example as shown in the view of FIG. **17E**, the second support portion **1142e** is of an equiaxed, or hexangular, configuration. In yet another example, as shown in the view of FIG. **17F**, the second support portion **1142f** may be angled to a general mid-plane **1105** of the elongated body **1102**. In yet another example as shown in the view of FIG. **17G**, the second support portion **1142g** is substantially co-planar yet oriented perpendicularly to the elongate body **1102**.

[0099] Although a specific wiring type has been disclosed earlier herein, such type is merely illustrative and hence, non-limiting of this disclosure. As shown best in the view of FIG. **18**, a multifunctional ceiling support assembly **1200** may be configured to include a connecting member **1232** having a second support portion **1242** that allows for a flexible PCB strip **1278** to be received therethrough. In fact, it is envisioned that such flexible PCB strip **1278** may be easier to route while also effectively catering to power distribution requirements and/or control requirements of elongated light assemblies (not shown) when multiple such ceiling support assemblies **1200** are for instance, daisy-chained or grouped together in another manner, or configuration commonly known in the art. In addition, the flexible PCB is thinner and therefore, also easier to route under a ceiling tile without significantly raising the ceiling tile above the T-Bar element, if any. By way of embodiments herein, it is hereby envisioned that the ceiling support assemblies of the present disclosure can serve to not only structurally form, in part, the ceiling grid of the suspended ceiling

arrangement but also function as part of an electrical circuit.

[0100] FIG. **19** is a bottom view of an embodiment suspended ceiling grid system layout wherein multifunctional ceiling support assemblies **1900** are mounted non-parallel to ceiling grid T-bars **1911** and **1913** which define x and y axes of the ceiling grid system respectively. In this embodiment, the ceiling tiles **1920** are cut to custom triangular shapes. Third connecting members **1960** (hidden from direct view and notated with dotted lines) extend up from the multifunctional ceiling support assemblies **1900** in a positive z direction and function to serve as spacer components that in combination with T-bars hold the ceiling tiles **1920** in position.

[0101] Modifications to embodiments of the present disclosure described in the foregoing are possible without departing from the scope of the present disclosure as defined by the accompanying claims. Expressions such as “including”, “comprising”, “incorporating”, “have”, “is” used to describe and claim the present disclosure are intended to be construed in a non-exclusive manner, namely allowing for items, components or elements not explicitly described also to be present. Reference to the singular is also to be construed to relate to the plural.

Claims

1. A multifunctional support assembly for use in a T-bar grid arrangement, the multifunctional lighting assembly comprising; a) an elongate body comprising an elongate planar base portion having first and second lateral ends, first and second longitudinal ends, a top plane, and a channel located in an interior space of a base portion bounded by first and second protrusions of the elongate body; b) first and second connecting members each positioned at the first and second longitudinal ends of the elongate body having support portions configured to be attached to the T-bar grid arrangement; wherein at least one connecting member comprises a support portion and a flexible PCB strip which allows for bridging of a T-bar of the T-bar grid arrangement to make electrical connections between multifunctional support assemblies without need of cutting or significant displacement of ceiling tiles of the T-bar grid arrangement.
2. The multifunctional support assembly of claim 1 wherein the T-bar grid arrangement is part of a suspended ceiling system comprising multiple ceiling panels.
3. The multifunctional support assembly of claim 2 which supports at least one ceiling panel.
4. The multifunctional support assembly of claim 2 configured to separate and position one or more of multiple ceiling panels within the T-bar grid arrangement.
5. The multifunctional support assembly of claim 2 wherein one or more of the multiple ceiling panels rests upon the top plane of the elongate body.
6. The multifunctional support assembly of claim 2, wherein the elongate body is at least partially recessed within tegular edges of two abutting ceiling panels of the multiple ceiling panels.
7. The multifunctional support assembly of claim 2 wherein the elongate body is positioned below a bottom surface of at least one of the multiple ceiling panels of the suspended ceiling system.
8. The multifunctional support assembly of claim 1 further comprising a third connecting member mounted on the top plane of the elongate body and extending up from the elongate body.
9. The multifunctional support assembly of claim 8 wherein the third connecting member comprises a first support portion coupled to the top plane of the elongate planar base portion and a second support portion extending vertically from the first support portion.
10. The multifunctional support assembly of claim 1 wherein a connecting member further comprises an anchoring hole that serves as an attachment point for suspension from a structural ceiling.
11. The multifunctional support assembly of claim 1, wherein a width of the elongate body is less than or equal to a width of a respective T-bar within the T-bar grid arrangement.
12. The multifunctional support assembly of claim 1 wherein an optical element redirects light emitted from a reflector or light guide.

- 13.** The multifunctional support assembly of claim 1 further comprising a component selected from a group consisting of light source, light emitting diode, alarm, sensor, audio sensor, acoustic sensor, motion sensor, position sensor, proximity sensor, light sensor, ultrasonic sensor, accelerometer, ventilation fan, heater, humidifier, electronic controller, power supply, battery, and wireless communication module.
- 14.** The multifunctional support assembly of claim 1 wherein the first or second connecting member is further comprised of a second support portion having a geometric configuration consisting of one of the following; quadrilateral, semi-hexagonal, semi-circular, equiaxed, and hexangular.
- 15.** The multifunctional support assembly of claim 1 further comprising an end cap positioned at the first or second longitudinal end of the elongate body.
- 16.** The multifunctional support assembly of claim 15 further comprising wiring routed through the end cap.
- 17.** The multifunctional support assembly of claim 1 wherein at least one of the group consisting of optical element, reflector, and printed circuit board is slidably mounted between the first and second protrusions of the elongate body.
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